

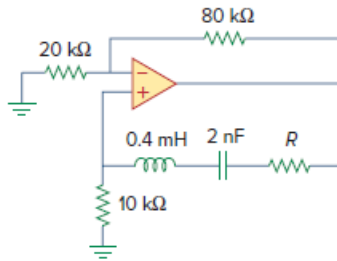
Problem

Consider the oscillator in Fig. 10.133.

(a) Determine the oscillation frequency.

(b) Obtain the minimum value of R for which oscillation takes place.

Figure 10.133:



Step-by-step solution

Step 1 of 6

(a)

Refer to circuit diagram in Figure 10.133 in the textbook.

Refer to section 10.9.2 in the textbook,

The feedback ratio is,

$$\frac{V_2}{V_o} = \frac{Z_p}{Z_s + Z_p}$$

Here,

$$Z_p = 10 \text{ k}\Omega$$

$$Z_s = j\omega L + \frac{1}{j\omega C} + R$$

$$= j\omega(0.4 \times 10^{-3}) + \frac{1}{j\omega(2 \times 10^{-9})} + R$$

Step 2 of 6

Calculate the feedback ratio.

$$\frac{V_2}{V_o} = \frac{Z_p}{Z_s + Z_p}$$

Substitute $10 \text{ k}\Omega$ for Z_p , and $j\omega(0.4 \times 10^{-3}) + \frac{1}{j\omega(2 \times 10^{-9})} + R$ for Z_s in the above equation.

$$\begin{aligned} \frac{V_2}{V_o} &= \frac{10 \text{ k}\Omega}{j\omega(0.4 \times 10^{-3}) + \frac{1}{j\omega(2 \times 10^{-9})} + R + 10 \text{ k}\Omega} \\ &= \frac{10 \text{ k}\Omega}{R + 10 \text{ k}\Omega + j\left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)} \\ &= \frac{(10 \text{ k}\Omega)\left(R + 10 \text{ k}\Omega - j\left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)\right)}{(R + 10 \text{ k}\Omega)^2 + \left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)^2} \\ &= \frac{(R + 10 \text{ k}\Omega)(10 \text{ k}\Omega) - j(10 \text{ k}\Omega)\left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)}{(R + 10 \text{ k}\Omega)^2 + \left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)^2} \end{aligned}$$

Step 3 of 6

To calculate the oscillation frequency, equate imaginary part of feedback ratio to zero.

$$\begin{aligned} \frac{(10 \text{ k}\Omega)\left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)}{(R + 10 \text{ k}\Omega)^2 + \left(\omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})}\right)^2} &= 0 \\ \omega(0.4 \times 10^{-3}) - \frac{1}{\omega(2 \times 10^{-9})} &= 0 \\ \omega^2 &= \frac{1}{(0.4 \times 10^{-3})(2 \times 10^{-9})} \\ \omega &= \sqrt{\frac{1}{0.8 \times 10^{-12}}} \\ &= \sqrt{1.25 \times 10^{12}} \\ &= 1.118 \text{ Mrad/sec} \end{aligned}$$

Step 4 of 6

Calculate the oscillation frequency in Hz.

$$f = \frac{\omega}{2\pi}$$

Substitute 1.118 Mrad/sec for ω in the above equation.

$$f = \frac{1.118 \text{ Mrad/sec}}{2\pi}$$

$$= 180 \text{ kHz}$$

Therefore, the value of oscillation frequency, f is **180 kHz**.

Step 5 of 6

(b)

Apply Kirchhoff's current law at inverting terminal.

$$\frac{V_2}{20 \text{ k}\Omega} + \frac{V_2 - V_o}{80 \text{ k}\Omega} = 0$$

$$\frac{V_o}{80 \text{ k}\Omega} = \left(\frac{1}{20 \text{ k}\Omega} + \frac{1}{80 \text{ k}\Omega} \right) V_2$$

$$\frac{V_o}{80 \text{ k}\Omega} = \left(\frac{4+1}{80 \text{ k}\Omega} \right) V_2$$

$$\frac{V_o}{V_2} = 5$$

Step 6 of 6

At oscillation frequency, the feedback ratio is,

$$\frac{V_2}{V_o} = \frac{10 \text{ k}\Omega}{j(1.118 \times 10^6)(0.4 \times 10^{-3}) + \frac{1}{j(1.118 \times 10^6)(2 \times 10^{-9})} + R + 10 \text{ k}\Omega}$$

$$= \frac{10 \text{ k}\Omega}{j447.2 + j447.22 + R + 10 \text{ k}\Omega}$$

$$= \frac{10 \text{ k}\Omega}{R + 10 \text{ k}\Omega}$$

Substitute $\frac{1}{5}$ for $\frac{V_2}{V_o}$ in the above equation.

$$\frac{1}{5} = \frac{10 \text{ k}\Omega}{R + 10 \text{ k}\Omega}$$

$$R + 10 \text{ k}\Omega = 50 \text{ k}\Omega$$

$$R = 40 \text{ k}\Omega$$

Therefore, the minimum value of resistance, R for which oscillation takes place is **40 kΩ**.